

ZEN SCHOOL OF MANAGEMENT USING CISCO PACKET TRACER AND NETWORKING CONCEPTS

Enterprise Network Design and Implementation Using Cisco Packet Tracer

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Project Description

The Zen School Management Network is a comprehensive enterprise-level network infrastructure designed and simulated using Cisco Packet Tracer. The project demonstrates the implementation of routing, switching, security, wireless networking, IoT integration, and network services within a school environment. The network connects multiple departments including administrative offices, staff rooms, laboratories, library, security room, AV room, server rooms, and guest lobby through a scalable and secure architecture.

The project was developed to simulate real-world networking scenarios and to gain practical experience in configuring enterprise networking technologies such as OSPF, DHCP, SSH, TACACS+, ACLs, NAT, FTP, WLAN, IoT, and NTP.

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1. INTRODUCTION

Modern educational institutions require a reliable, secure, and scalable network infrastructure to support communication, resource sharing, administration, and learning activities. The Zen School Management Network was designed to fulfill these requirements by creating a simulated enterprise network using Cisco Packet Tracer.

The network integrates multiple technologies that are commonly deployed in real-world organizations, providing practical exposure to enterprise network design and implementation.

2. PROJECT OBJECTIVES

The primary objectives of this project are:

- Design an enterprise-level school network architecture.
- Implement dynamic routing using OSPF.
- Configure DHCP for automatic IP allocation.
- Provide secure remote access through SSH.
- Implement centralized authentication using TACACS+.
- Secure network resources using Access Control Lists.
- Configure Network Address Translation (NAT).
- Enable file sharing using FTP services.
- Integrate IoT devices into the network.
- Deploy wireless networking infrastructure.

- Synchronize devices using Network Time Protocol (NTP).
- Verify end-to-end connectivity across all departments.

3. NETWORK OVERVIEW

The network consists of multiple departments connected through Cisco ISR4331 routers and Cisco Catalyst switches.

Departments included in the network are:

- Server Room
- Office
- Staff Room
- Library
- Security Room
- Lab 1
- Lab 2
- Lab 3
- AV Room
- Guest Lobby

The network also includes:

- Wireless LAN Controller (WLC)
- Lightweight Access Point
- Wireless Clients
- IoT Devices
- Servers
- Personal Computers

The routers are interconnected through WAN links and exchange routing information dynamically using OSPF.

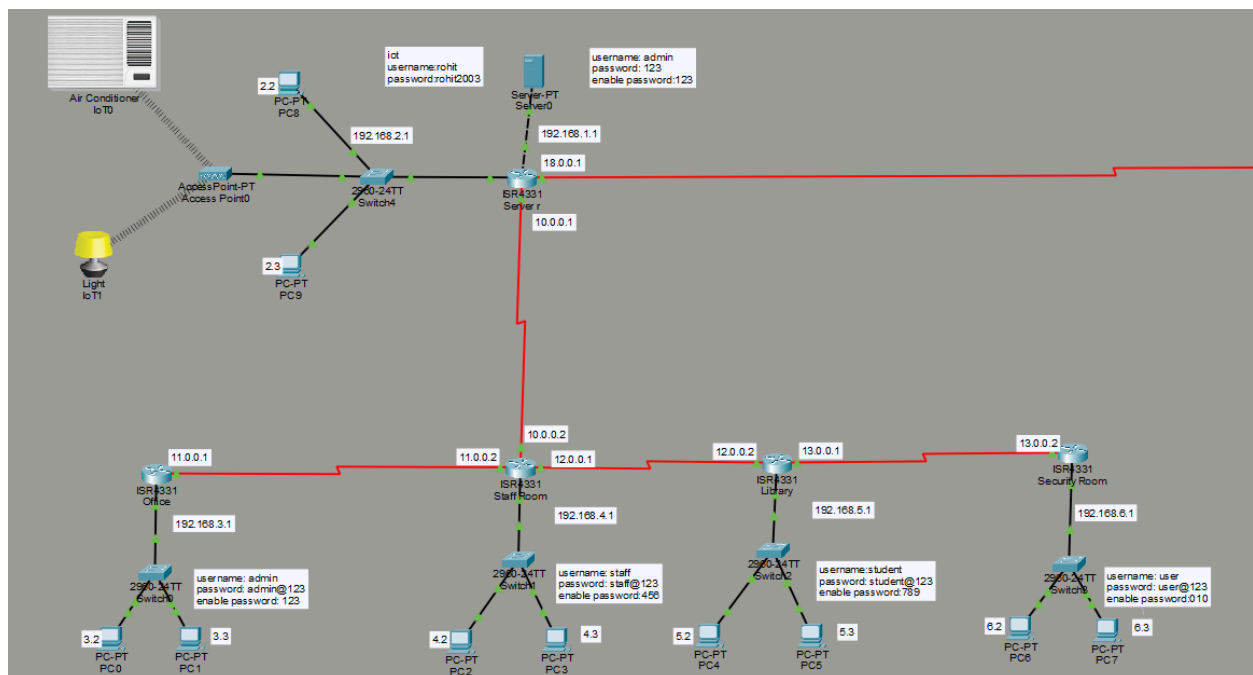
4. NETWORK TOPOLOGY

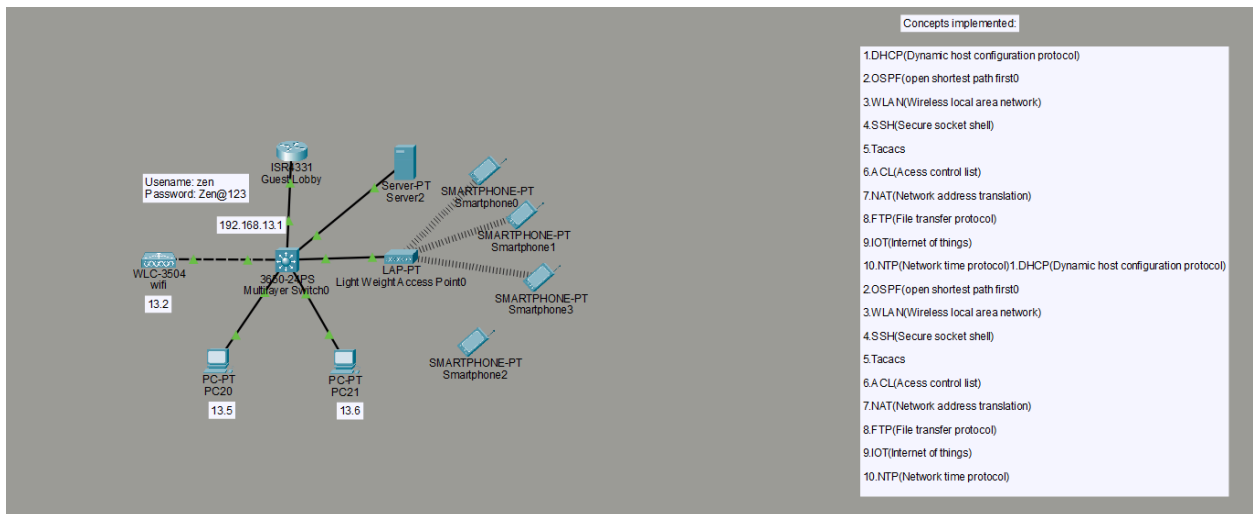
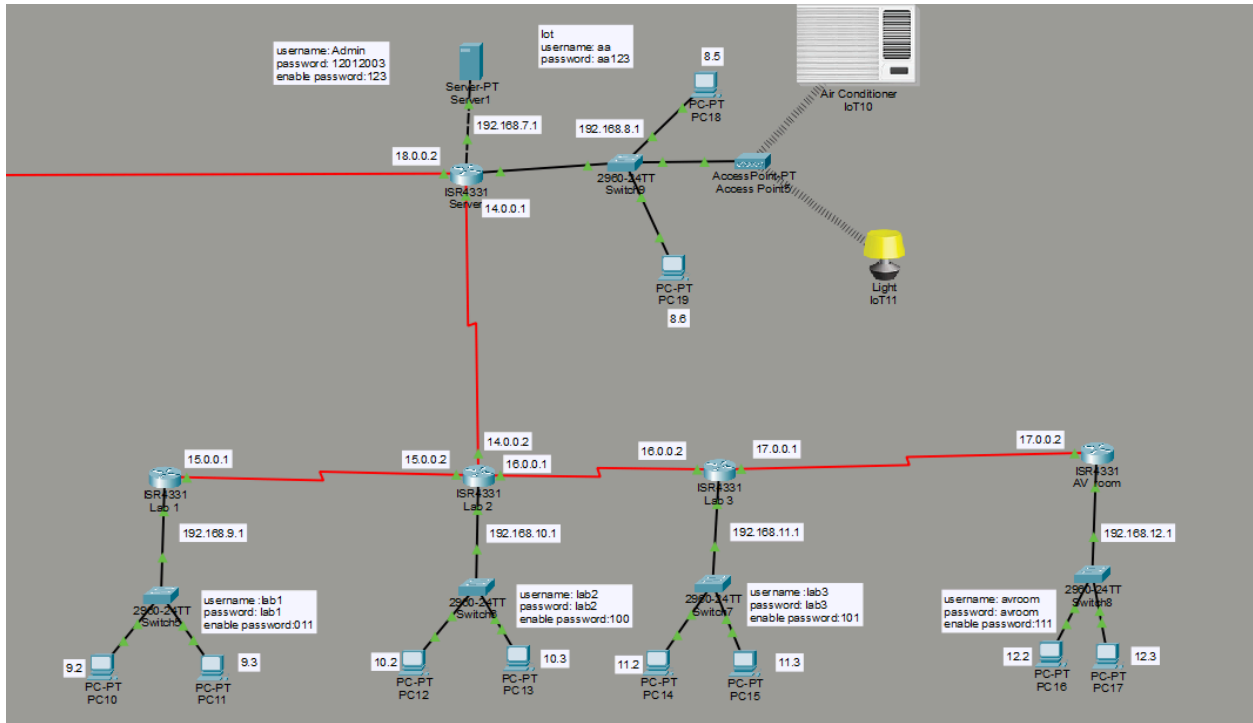
The topology is designed using a hierarchical enterprise architecture.

The infrastructure includes:

- Core routing layer
- Departmental access layer
- Wireless network segment
- Server segment
- IoT segment
- Guest network segment

The network provides communication between all departments while maintaining security and efficient routing.





5. IP ADDRESSING SCHEME

The network uses private IPv4 addressing.

Department Networks:

Server Room – 192.168.1.0/24

IoT Room – 192.168.2.0/24

Office – 192.168.3.0/24

Staff Room – 192.168.4.0/24

Library – 192.168.5.0/24

Security Room – 192.168.6.0/24

Server Segment 2 – 192.168.7.0/24

Wireless Network – 192.168.8.0/24

Lab 1 – 192.168.9.0/24

Lab 2 – 192.168.10.0/24

Lab 3 – 192.168.11.0/24

AV Room – 192.168.12.0/24

Guest Lobby – 192.168.13.0/24

WAN Networks:

10.0.0.0/30

11.0.0.0/30

12.0.0.0/30

13.0.0.0/30

14.0.0.0/30

15.0.0.0/30

16.0.0.0/30

17.0.0.0/30

18.0.0.0/30

6. DEVICES USED

Networking Devices:

- Cisco ISR4331 Routers
- Cisco Catalyst 2960 Switches
- Cisco Catalyst 3560 Multilayer Switch
- Wireless LAN Controller (WLC)
- Lightweight Access Point

End Devices:

- Personal Computers
- Smartphones
- Servers

IoT Devices:

- Smart Air Conditioner
- Smart Light

7. TECHNOLOGIES IMPLEMENTED

The following networking technologies were implemented:

1. DHCP (Dynamic Host Configuration Protocol)
2. OSPF (Open Shortest Path First)
3. WLAN (Wireless Local Area Network)
4. SSH (Secure Shell)
5. TACACS+
6. Access Control Lists (ACL)
7. Network Address Translation (NAT)
8. FTP (File Transfer Protocol)
9. Internet of Things (IoT)
10. Network Time Protocol (NTP)

8. OSPF ROUTING CONFIGURATION

Open Shortest Path First (OSPF) was configured on all routers to enable dynamic route exchange throughout the network.

Benefits:

- Automatic route learning
- Fast convergence
- Scalability
- Efficient routing

All routers successfully established OSPF neighbor relationships and exchanged routing information.

9. DHCP CONFIGURATION

DHCP was implemented to automate IP address assignment for client devices.

The DHCP service provides:

- IP Address
- Subnet Mask
- Default Gateway

Benefits:

- Reduced manual configuration
- Simplified network management
- Faster deployment

10. WLAN IMPLEMENTATION

Wireless connectivity was implemented using a Wireless LAN Controller and Lightweight Access Point.

Wireless users can access network resources without physical cabling.

Benefits:

- Mobility
- Centralized wireless management
- Flexible connectivity

11. SSH CONFIGURATION

SSH was configured on routers to provide secure remote administration.

Benefits:

- Encrypted communication
- Secure device management
- Protection against unauthorized access

12. TACACS+ AUTHENTICATION

TACACS+ was configured to provide centralized authentication services.

Benefits:

- Centralized user management
- Enhanced security
- User-based authentication and authorization

13. ACCESS CONTROL LISTS (ACL)

ACLs were implemented to control access between network segments.

Functions:

- Permit authorized traffic
- Deny unauthorized access
- Protect sensitive resources

Benefits:

- Improved security
- Better traffic control

14. NETWORK ADDRESS TRANSLATION (NAT)

NAT was configured to translate private IP addresses and facilitate communication between different network segments.

Benefits:

- Address conservation
- Enhanced security
- Simplified network administration

15. FTP SERVER CONFIGURATION

FTP services were configured on the server for file sharing and resource distribution.

Features:

- File upload
- File download
- Centralized storage

Benefits:

- Efficient resource sharing
- Simplified file management

16. IoT INTEGRATION

IoT devices were integrated into the network to simulate a smart campus environment.

Devices:

- Smart Air Conditioner
- Smart Light

Benefits:

- Device automation
- Remote management
- Smart infrastructure implementation

17. NTP CONFIGURATION

Network Time Protocol (NTP) was configured to synchronize time across network devices.

Benefits:

- Accurate timestamps
- Consistent logging
- Easier troubleshooting

18. TESTING AND VERIFICATION

The following tests were successfully performed:

Connectivity Testing:

- Ping between departments
- End-to-end communication verification

Routing Verification:

- OSPF neighbor validation
- Route table verification

DHCP Testing:

- Dynamic IP assignment confirmed

SSH Testing:

- Successful remote access

WLAN Testing:

- Wireless connectivity verified

FTP Testing:

- Successful file transfer

IoT Testing:

- Device communication verified

19. RESULTS

The project successfully achieved the following outcomes:

- ✓ Dynamic Routing Using OSPF
- ✓ Automated IP Address Allocation

- ✓ Secure Remote Administration
- ✓ Centralized Authentication
- ✓ Wireless Network Deployment
- ✓ Access Control Implementation
- ✓ NAT Configuration
- ✓ FTP Services
- ✓ IoT Integration
- ✓ NTP Synchronization
- ✓ End-to-End Connectivity
- ✓ Enterprise Network Security

20. CONCLUSION

The Zen School Management Network successfully demonstrates the design and implementation of a secure, scalable, and efficient enterprise network using Cisco Packet Tracer. The project integrates routing, switching, wireless networking, security mechanisms, network services, and IoT technologies to create a realistic educational institution network environment.

The implementation of OSPF, DHCP, SSH, TACACS+, ACLs, NAT, FTP, WLAN, IoT, and NTP provides practical exposure to enterprise networking concepts and showcases the skills required for real-world network administration and management. The project serves as an effective demonstration of CCNA-level networking knowledge and hands-on configuration experience.